INCH-POUND

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MILITARY SPECIFICATION

PROJECTILES, 155MM, M483A1 - M718A1/M741A1 - M692/M731 METAL PARTS ASSEMBLY

This Specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions, and packaging for Projectiles, 155MM, M483A1 - M718A1/M741A1 - M692/M731 Metal Parts (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Method and Pre-treatment of Ferrous Surfaces for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, NJ 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1320

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MILITARY

MIL-C-5541	- Chemical Conversion Coatings on Aluminum Alloys
MIL-M-6867	- Magnetic Inspection Units
MIL-A-8625	- Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-410	- Nondestructive Testing Personnel Qualification and Certification
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-1235	- Single- and multi-level Continuous Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-1949	- Inspection, Magnetic Particle

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: DODSSP Customer Service, Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.16)

U.S. Army Armament Research, Development and Engineering Center (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

8837839	- Pallet for Projectile, 155MM, HE and GB Hollow Base Type
10542902-1	- Projectile, 155MM, M483A1, Metal Parts Assy
10542902-2	- Projectile, 155MM, M718A1/M741A1, Metal Parts Assy
10542902-3	- Projectile, 155MM, M692/M731, Metal Parts Assy

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INSPECTION EQUIPMENT DRAWINGS

9280318	- Magnetic, Particle Standard for M483A1 Body
9280388	- Eddy Current Standard for 155MM, HE, M718A1/M741A1 Base
9280392	- Eddy Current Equipment for 155MM, HE, M483, M692/M731 Base
9280400	- Eddy Current Equipment for 155MM, M483, M718A1/M741A1, M692/M731 Ogives
9280401	- Eddy Current Standard for 155MM, M483A1, M692/M731 and M718A1/M741A1 Ogives
9280404	- Eddy Current Standard for 155MM, M483, M692/M731 Base
9280406	- Electrical Conductivity Measurement System 155MM, M483A1, M692/M731, M718A1/M741A1 Base

(Copies of other Government documents, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Method E8	- Tension Testing of Metallic Materials
ASTM Method E10	- Brinell Hardness of Metallic Materials
ASTM Method E18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM Method G47	- Determining Susceptibility to Stress-Corrosion Cracking of High-Strength Aluminum Alloy Products
ASTM Method E103	- Rapid Indentation Hardness Testing of Metallic Materials
ASTM Method B117	- Salt Spray (Fog) Testing
ASTM Method B247	- Aluminum and Aluminum Alloy Die, Hand and Rolled Ring Forgings
ASTM Method E1004	- Electromagnetic (Eddy Current) Measurements of Electrical Conductivity

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

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SOCIETY OF AUTOMOTIVE ENGINEERS

AMS 2375 - Specification Control of Forgings Requiring First Article Approval

(Applications for copies should be addressed to Society of Automotive Engineers, 400, Commonwealth Drive, Warrandale, PA 15906.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials, parts and assemblies shall comply with the requirements specified on the applicable drawings, referenced specifications.

3.2 Projectile body.

3.2.1 Forming. When a cold working operation is used in forming the projectile body, the maximum reduction in cross sectional area shall not exceed 22 percent. Process drawings showing the prior to cold work and after cold work process piece dimensions shall be submitted to cognizant engineering agency, for approval prior to use. Once approved, the process drawings submitted, shall not be changed without prior approval of the procuring activity. The hardness of the process piece prior to a cold working operation shall not exceed 241 Brinell (see 6.7).

3.2.2 Stress relief. Prior to the band welding operation, projectile bodies which have been cold worked shall be stress relieved at 1000°F minimum temperature (body) for a minimum time of 1/2 hour.

3.2.3 Heat treatment. Each projectile body (after application of the welded overlay band) shall be austenitized, quenched and tempered uniformly to meet the mechanical properties specified on the applicable drawing. No work involving plastic flow is permitted after tempering unless followed by a stress relief. The contractor shall submit, for information only, the heat treatment procedure to the technical agency.

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3.3 Rotating band.

3.3.1 Rotating band application. The rotating band shall be overlay welded using the gas metal arc welding process in accordance with the applicable drawings and specifications. Gas metal arc welding process in this instance means the use of a consumable electrode with a non-electrode auxiliary wire. The width of the band shall be obtained by oscillating the welding head and the auxiliary wire feed system. The process for applying the rotating band, once established, must be maintained, controlled and monitored by the contractor. Procedures for maintaining, controlling and monitoring the process must be submitted to the technical agency for approval prior to the start of production. Documented records of the process control must be made available to the Government upon request (see 6.10).

3.3.1.1 Minimum melting of steel. Upon removal of the rotating band material, the steel band seat shall demonstrate visual evidence of minimum melting due to gilding metal adhesion during application of the rotating band. Visual comparison standards will be provided by the Government.

3.3.1.2 Subsurface defects. There shall be no evidence of either intergranular penetration of gilding metal or weld start-stop cratering when the body band seat is turned down to 5.875 plus or minus 0.010 inch diameter and tested as specified in 4.5.3.1.

3.3.2 Rotating band seat. Body material (steel) under the rotating band shall not be visible in the cannelure diameter.

3.3.2.1 Process control. Body material (steel) under the rotating band shall not extend above the finished bourrelet surface. A process requirement that will control this feature is depicted on applicable body dwg. (see 4.4.2.6 - Dwg. No.). The contractor may, at his option, use a control other than the one defined on the drawing. The manufacturing procedure and the inspection plan for controlling process shall be forwarded to the responsible technical agency for approval prior to its implementation into production.

3.3.3 Foreign matter on band seat. Prior to banding, the band seat should be free from any contaminants that would be detrimental to the adhesion of the band to the band seat.

3.4 Torque.

3.4.1 Fiberglass wrapping to body. The fiberglass wrapping shall withstand a minimum torque of 550 foot pounds.

3.5 Metal defects.

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3.5.1 Ogive, body and base. The ogive, body and base shall be free from all metal defects, in accordance with the defect standards (see 3.5.3) including cracks, splits, bursts, cold shuts, pipes, porosity, inclusions, folds, seams and other metal defects. In addition, the cavity of the components shall be free of scale, fins, burrs, draw marks, laminations, imbedded foreign matter, and pits (see 3.5.3).

3.5.2 Rotating band. The rotating band and band seat areas shall have no cracks, lack of bonding, or other metal defects in accordance with defect standards (see 3.5.3).

3.5.3 Defect standards. Defect standards must be approved by the technical engineering agency. All visual standards for defects will be established for each contract after the first month's production or whenever requested by the procuring contracting agency.

3.6 Proving ground function test. The assembled projectile metal parts shall reveal no evidence of failure when tested. Metal parts failure shall be:

a. Loss or breakup of metal part components in flight (except obturator).

b. Shear of the rotating band as evidenced by complete lack of engraving.

c. Loss or loose fiberglass wrapping.

d. Base plug and cargo not ejecting (except excess pressure). (Failure analysis must determine the cause to be the projectile metal parts.)

3.7 Ogive and base.

3.7.1 Microstructure. The microstructure of the ogive and base shall comply with the microstructure depicted in Figures 1a, 1b or 2. Two examples of acceptable microstructure are shown in Figures 1a and 1b. In Figure 1a, the insoluble phase is well broken up, randomly distributed within a matrix which is essentially unrecrystallized. In Figure 1b, the insoluble phase is well broken up and randomly distributed with a matrix which is partially recrystallized. An example of a marginally acceptable microstructure is shown in Figure 2. In this Figure, the insoluble second phase exhibits some discontinuity and appears to be elongated. An example of an unacceptable microstructure is shown in Figure 3. The insoluble phase is distributed in a more continuous form (network) within a matrix which shows a large degree of microsegragation (or coring). The microstructure in Figure 3 must be avoided because it is detrimental to the integrity of the material (see 6.12).

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3.7.2 Heat treatment. Each base and ogive shall be solution heat treated and aged uniformly to meet the mechanical properties specified on the applicable drawing.

3.7.2.1 Base heat treatment verification. Each base shall be Rockwell B82 minimum.

3.7.3 Monitoring of hot forge temperature. When bases and ogives are hot forged, the forger shall provide continuous, automatic monitoring of slug temperature prior to hot forging. Details shall be provided in accordance with 3.8.1.

3.7.4 Storage. Heat treated forgings, semi finished, or finished components shall be stored indoors. Forgings shall be processed into finished components in a sequential forging lot basis (i.e., first in, first out).

3.8 First article. When specified in the contract or purchase order (see 6.3), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.8.1 First article approval of aluminum forgings. First article approval of forgings, forging drawings, process, microstructure, and controls to be used in producing subsequent production forgings shall be obtained from the Government technical agency prior to initiating production. The procedures for obtaining first article approval shall be in accordance with AMS 2375.

3.9 Control dimensions.

3.9.1 Tool control dimensions. Dimensions marked tool control shall be gaged during First Article Testing and after eight hours maximum production on each machine in operation and prior to any tooling change. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided the contractor has established correlation between the tool dimensions and the components dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor.

3.9.2 Process control dimensions. The contractor shall prepare a process control plan for the dimensions marked process control. The plan shall be submitted to the technical agency for approval prior to use. Dimensions to be gaged for process control are to be assured to a Level III inspection conformance criteria.

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3.10 Protective finish. Details of the proposed procedure for zinc phosphate coating, hard anodic coating, anodic coating (with duplex seal), and chemical conversion coating to be used by the contractor shall be submitted in writing to the Government Technical Agency and written approval received prior to initiation of production. The proposed procedure shall include materials, process and equipment to be used by the contractor, detailed method of control including limits for time, temperature, concentration and all other information pertinent to the processes. Approval of process, materials, or equipment implies no guarantee of acceptance of the results obtained in use.

3.10.1 Ogive. First article ogives shall withstand the salt spray test as prescribed in 3.10.1.2 of MIL-A-8625. Production lot ogives shall comply with 3.10.1.2 of MIL-A-8625 except withstand 72 hours and show no more than a total of 3 corrosion spots or pits on the significant surface.

3.10.2 Base. After exposure to a 72 hours salt spray, the chemical conversion coating on the base shall not exhibit corrosion spots or pits on more than 5 percent of the test surface defined in 4.5.8.1.4.

3.11 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following.

3.11.1 Burr and sharp edges. No component shall have a burr or sharp edge which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.11.2 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease, oil, rust, or other foreign matter which may interfere with final inspection and completely free of same prior to shipping.

3.11.3 Threads. Threads shall be full and undamaged, for the entire minimum length or depth as required by the applicable drawings.

3.11.4 Painting. Painting shall be complete, void free, and undamaged.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-A-48078 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

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PART DESCRIPTION	DRAWINGS			QUANTITY	
	M483A1	M718A1/M741A1	M692/M731		
Projectile Metal Parts Assembly Complete with Bases	10542902-1	10542902-2	10542902-3	25	
Projectile Subassembly	10542910-1	10542910-2	9347119	5	
Body wrapped	10542911-1	10542911-2	10542911-3	5	
Body	10542912	9278078	9298354	5	
Body (prior to banding)	10542912	9278078	9298354	2	
Ogive	10542916	10542916	9298324	5	
Base (prior to hardcoat and drilling wrench holes)	10542904	9277913	9298335	5	
Base Hardcoated	10542904	9277913	9298335	5	(*10)
Obturator	10542907	10542907	10542907	5	

* If the optional nickel acetate seal is utilized.

Twenty of the above projectile metal parts assemblies and bases shall be forwarded to a designated load plant for loading and shall then be forwarded to a Government proving ground for ballistic testing.

4.3.2 Inspection to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification (see Table I) and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

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TABLE I. First article inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile Metal Parts for Projectile, 155MM, HE: M483A1, HE: M718A1/M741A1 or HE: M692/M731	SHEET 1 OF 3	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Ogive M483A1/M718A1/M741A1 (Dwg 10542916) 1/ M692/M731 (Dwg. 9298324) 1/			
	Examination for defects	5, 0-1	3.1	4.4.2.1
	Salt spray test	1, 0-1	3.1	4.5.8.1/4.5.8.1.1
	Yield strength & elongation	2Specimens, 0-1	3.7.2	4.5.1.5
	Eddy Current Inspection	5, 0-1	3.5.1	4.5.11.2
	Obturator (Dwg 10542907)			
	Examination for defects	5, 0-1	3.1	4.4.2.3

NOTES:
1. See para. 3.8.1 herein.

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TABLE I. First article inspection
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile Metal Parts for Projectile, 155MM, HE: M483A1, HE: M718A1/M741A1 or HE: M692/M731	SHEET 2 OF 3	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Base			
	M483A1 (Dwg 10542904) 1/			
	M718A1/M741A1 (Dwg 9277913) 1/			
	M692/M731 (Dwg 9298335) 1/			
	Examination for defects	5, 0-1	3.1	4.4.2.4
	Yield strength & percent elongation	2 Specimens	3.1	4.5.1.5
	Salt spray test (hard anodic coating)	1, 0-1	3.1	4.5.8.1/4.5.8.1.2
	Salt spray test (nickel acetate seal)	5, 0-1	3.1	4.5.8.1/4.5.8.1.3
	Salt spray test (chemical conversion coating)	1, 0-1	3.10.2	4.5.8.1/4.5.8.1.4
	Hardcoat thickness test	1, 0-1	3.1	4.5.8.2
	Eddie Current Inspection	5, 0-1	3.5.1	4.5.11.1
	Base Forging			
	M483A1 (Dwg 10542904) 1/			
	M692/M731 (Dwg 9298335) 1/			
	M718A1/M741A1 (Dwg 9277913) 1/			
	Stress corrosion cracking test 2/	1, 0-1	3.1	4.5.1.7
	Body			
	M483A1 (Dwg 10542912)			
	M718A1/M741A1 (Dwg 9278078)			
	M692/M731 (Dwg 9298354)			
	Examination for defects	5, 0-1	3.1	4.4.2.6
	Yield strength & percent elongation	2, 0-1	3.2.3	4.5.1.4

NOTES:
1. See para 3.8.1 herein.
2. A stress corrosion crack test will be performed for first article inspection or if a different aluminum supplier is used.

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TABLE I. First article inspection
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE Projectile Metal Parts for Projectile, 155MM, HE: M483A1, HE: M718A1/M741A1 or HE: M692/M731	SHEET 3 OF 3	DRAWING NUMBER See Below NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
	Body wrapped			
	M483A1 (Dwg 10542911-1)			
	M718A1/M741A1 (Dwg 10542911-2)			
	M692/M731 (Dwg 10542911-3)			
	Examination for defects	5, 0-1	3.1	4.4.2.6
	Torque test of fiberglass wrap	2, 0-1	3.4	4.5.9
	Fiberglass content	2, 0-1	3.1	4.5.10
	Projectile subassembly (before paint)			
	M483A1 (Dwg 10542910-1) 1/			
	M718A1/M741A1 (Dwg 10542910-2) 1/			
	M692/M731 (Dwg 10542910-3) 1/			
	Examination for defects	5, 0-1	3.1	4.4.2.7
	Projectile subassembly (painted)			
	M483A1 (Dwg 10542902-1) 1/			
	M718A1/M741A1 (Dwg 10542902-2) 1/			
	M692/M731 (Dwg 10542902-3) 1/			
	Examination for defects	25, 0-1	3.1	4.4.2.8
	Salt spray test	2, 0-1	3.1	4.5.8.1
	Ballistic tests		3.6	4.5.12
	A. Excess pressure	10, 0-1	3.6	4.5.12.1
	B. Zone 3	5, 0-1	3.6	4.5.12.2
	C. Service pressure	5, 0-1	3.6	4.5.12.3

NOTES:
1. See para 3.8.1 herein.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements. In addition, each projectile inspection lot formation shall comply with the following:

a. Ogives from not more than one forging manufacturer.

b. Bases from not more than one forging manufacturer.

c. Bodies from not more than one forging manufacturer.

4.4.1.1 Body heat treatment lot. A heat treatment lot of bodies is defined as follows:

4.4.1.1.1 Standard procedure. A heat treatment lot of bodies shall consist of bodies from only one mill heat of steel that have been heat treated in the same heat treatment equipment, at the same temperature and for the same time cycles, in one unchanged process. The maximum lot size shall be 2000 bodies. If a mill heat exceeds 2000 bodies, the mill heat shall be divided into any convenient number of heat treatment lots that would preclude the need for forming miscellaneous heat treatment lots.

4.4.1.1.1.1 Miscellaneous lot. Bodies manufactured from small heats of steel, or bodies that have become separated from previously delivered lots may be grouped into a miscellaneous heat treatment lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.1.1. The maximum lot size shall be 2000 bodies.

4.4.1.1.2 Use. The heat treatment lot constituted per 4.4.1.1.1, 4.4.1.1.1.1 shall be the basis for selecting samples for mechanical properties tests. Bodies in each heat treatment lot shall be identified to that particular lot until satisfactory completion of mechanical properties tests.

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4.4.1.2 Heat treatment lot (ogive or base). A heat treatment lot of ogives and bases shall consist of components manufactured from material of the same alloy and temper from one manufacturer. At no time will the lot quantity exceed 4800. A heat treatment lot consisting of single component configuration shall be heat treated by the same continuous process, in the same equipment, and using the same temperature and time cycles. A heat treatment lot shall be considered to have been processed continuously if the parts are solutionized and quenched in an uninterrupted manner with no more than a ten (10) hour delay for non-equipment related shut down(s) provided normal furnace operation temperatures are maintained. If the heat treat process is interrupted to process different components, then different lot numbers must be assigned to the parts produced before and after the interruption. Each heat treat lot shall be aged in the same furnace at the same time.

The heat treatment lot thus constituted shall be the basis for selecting samples for mechanical properties tests. Components in each heat treatment lot shall remain segregated and identified until satisfactory completion of mechanical properties tests.

4.4.1.3 Miscellaneous lot (ogive or base). Components manufactured from small lots of material or those that have become separated from previously heat treated lots may be grouped into a miscellaneous lot. Such a lot shall be identified by a miscellaneous lot code number and processed in accordance with 4.4.1.2. The lot thus constituted shall be the basis for selecting samples for the mechanical properties tests.

4.4.1.4 Projectile lot. Heat treatment lots that have been determined to have complied with the yield strength and percent elongation requirements of the applicable drawing may be regrouped for assembly into projectile lots.

4.4.2 Examinations and tests.

a. Classification of characteristics. Quality conformance examinations and tests are specified in the following Classification of Characteristics paragraphs. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing and specification requirements utilizing as a minimum the conformance criteria specified. When cited herein, attributes sampling inspection shall be conducted in accordance with table II below, using the inspection levels stated in the Classification of Characteristics paragraphs.

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TABLE II. Attributes sampling inspection

Lot Size	Inspection Levels I	III	V
2 to 8	*	*	5
9 to 15	*	*	5
16 to 25	*	*	5
26 to 50	*	32	5
51 to 90	*	32	13
91 to 150	*	32	13
151 to 280	*	32	20
281 to 500	*	32	20
501 to 1200	*	80	20
1201 to 3200	1250	80	32
3201 to 10000	1250	125	32
10001 to 35000	1250	125	50
35001 to 150000	1250	125	50
150001 to 500000	1250	200	50
5000001 and over	1250	200	50

Numbers under inspection levels indicate sample size; asterisks (*) indicate one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection. Accept on zero and reject on one or more for all inspection levels.

b. Alternative quality conformance provisions. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.17). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provision(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

c. Automatic gaging. Facilities equipped with Government furnished automatic gaging capable of performing 100 percent inspection of any characteristic shall utilize this capability in accordance with contract.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE Ogive (Before anodizing)	SHEET 1 OF 3	DRAWING NUMBER 10542916 or 9298324 as applicable NEXT HIGHER ASSEMBLY 10542910-1,-2 or9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Metal defective	100%	3.5	4.5.11.2
b	Hydrostatic 2/		3.1	4.5.5
Major				
101	Pitch diameter of external thread, min	Level III	3.1	Gage
102	Major diameter of external thread, min	Level III	3.1	Gage
103	Diameter of cavity, rear	Level III	3.1	Gage
104	Length from rear face to forward inside basic diameter	Level III	3.1	Gage
105	Length from rear face to rear inside basic diameter	Level III	3.1	Gage
106	Variation in wall thickness	Level III	3.1	Gage
107	Pitch diameter of fuze thread, max	Level III	3.1	Gage
108	Minor diameter of fuze thread, max	Level III	3.1	Gage
109	Second diameter rear of fuze thread undercut	Level III	3.1	Gage
110	First diameter rear of fuze thread undercut	Level III	3.1	Gage

NOTES:
2. Sample size as specified in applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE Ogive (Before anodizing)	SHEET 2 OF 3	DRAWING NUMBER 10542916 or 9298324 (as applicable)
			NEXT HIGHER ASSEMBLY 10542910-1,-2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
111	Depth from front face to start of second diameter rear of fuze thread undercut	Level III	3.1	Gage
112	Length from front face to forward basic O.D.	Level III	3.1	Gage
113	Length from front face to rear basic O.D.	Level III	3.1	Gage
114	Length of fuze thread, min	Level III	3.1	Gage
115	True position of second diameter rear of fuze thread undercut with pitch diameter of fuze thread	Level III	3.1	Gage
116	True position at rear of fuze thread undercut with pitch diameter of fuze thread	Level III	3.1	Gage
117	Depth from rear face to bottom of cavity	Level III	3.1	Gage
118	Runout of outer ogive surface to body thread pitch diameter and shoulder (2 places) 1/	Level III	3.1	Gage

NOTES:
1. These inspections are not required when components are to be supplied as assemblies.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ogive (Before anodizing)	3 OF 3	10542916 or 9298324 (as applicable)
			NEXT HIGHER ASSEMBLY: 10542910-1,-2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
119	Runout of projected fuze thread pitch diameter (2 places) to body thread pitch diameter, body shoulder and fuze mating surface 1/	Level III	3.1	Gage
120	Yield strength and percent elongation 2/		3.7.2	4.5.1.5
121	Diameter of lock ring groove 3/	Level III	3.1	Gage
122	Parallelism of rear face with body/ogive shoulder 3/	Level III	3.1	Gage
123	Microstructure 2/		3.7.1	4.5.6
Minor				
201	Diameter of body thread undercut	Level V	3.1	Gage
202	Length from front face to shoulder	Level V	3.1	Gage
203	Depth from front face to rear of fuze thread undercut	Level V	3.1	Gage
204	Diameter of fuze thread undercut	Level V	3.1	Gage
205	Length, total	Level V	3.1	Gage
206	Radii or chamfers missing or incorrect	Level V	3.1	Visual
207	Surface finish improper	Level V	3.1	Visual
208	Burr	Level V	3.11	Visual
209	Foreign matter	Level V	3.11	Visual
210	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:
1. These inspections are not required when components are to be supplied as assemblies.
2. Sample size as specified in applicable test method paragraph.
3. Applicable to M692/M731 only.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Ogive (after anodizing)	1 OF 1	10542916 or 9298324 (as applicable)
			NEXT HIGHER ASSEMBLY: 10542916-1,-2 or 9348324 (as applicable)

CLASSIFICATION		EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical		None defined			
Major					
	101	Pitch diameter of external thread, max 2/	Level III	3.1	Gage
	102	Major diameter of external thread, max 2/	Level III	3.1	Gage
	103	Pitch diameter of fuze thread, min	Level III	3.1	Gage
	104	Minor diameter of fuze thread, min	Level III	3.1	Gage
	105	Salt spray 1/		3.1	4.5.8.1/4.5.8.1.1
Minor					
	201	Foreign matter	Level V	3.11	Visual
	202	Workmanship	Level V	3.11	Visual

NOTES:
1. Sample size as specified in applicable test method paragraph.
2. These inspections are not required when components are to be supplied as assemblies.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Obturator	1 OF 1	10542907
			NEXT HIGHER ASSEMBLY: 10542902-1, -2 or -3 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	True position of rear I.D. with forward I.D.	Level III	3.1	Gage
102	Diameter outside, front	Level III	3.1	Gage
103	Diameter outside, rear	Level III	3.1	Gage
104	Diameter inside, front	Level III	3.1	Gage
105	Depth of forward inside diameter	Level III	3.1	Gage
106	Thickness of wall at rear	Level III	3.1	Gage
107	Length, total	Level III	3.1	Gage
Minor				
201	Radii or chamfers missing or incorrect	Level V	3.1	Visual
202	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Base	SHEET 1 OF 3	9298335, 9277913 or 10542904 (as applicable)
			NEXT HIGHER ASSEMBLY: 10542902-1, -2 or -3 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Metal defective	100%	3.5	4.5.11.1
Special				
a	Barespot in anodic hardcoat	100%	3.1	Visual
b	Heat treatment verification stamp missing (indicating failure to pass test)	100%	3.1	Visual
Major				
101	Runout forward of taper with pitch diameter of threads and shoulder 3/	Level III	3.1	Gage
102	Runout rear of taper with pitch diameter of threads and shoulder 3/	Level III	3.1	Gage
103	Straightness of a diametral element 3/	Level III	3.1	Gage
104	Diameter of cavity 2/	Level III	3.1	Gage
105	Length from shoulder to forward basic diameter	Level III	3.1	Gage
106	Length from shoulder to rear basic diameter	Level III	3.1	Gage
107	Length from front face to shoulder	Level III	3.1	Gage
108	Length of thread 3/	Level III	3.1	Gage

NOTES:
2. Applies to M483-M692/M731 only.
3. Inspect prior to application of chemical conversion coating.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.4	TITLE Base	SHEET 2 OF 3	DRAWING NUMBER 9298335, 9277913 or 10542904 (as applicable) NEXT HIGHER ASSEMBLY 10542902-1, -2 or -3 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
109	Thickness through base web	Level III	3.1	Gage
110	Pitch diameter of thread 3/	Level III	3.1	Gage
111	Major diameter of thread, min 3/	Level III	3.1	Gage
112	Thread(s) damaged or not full	Level III	3.1	Visual
113	Yield strength and percent elongation 1/		3.7.2	4.5.1.5
114	Salt spray (nickel acetate seal) 1/		3.1	4.5.8.1/4.5.8.1.3
115	Flatness of front face 4/	Level III	3.1	Gage
116	Parallelism of front face with shoulder 4/	Level III	3.1	Gage
117	Inside diameter of spherical cutout 5/	Level III	3.1	Gage
118	True position of spherical cutout with pitch diameter of threads 5/	Level III	3.1	Gage
119	Depth from forward face to bottom of center of spherical cutout 5/	Level III	3.1	Gage
120	Salt spray (hard anodic coating) 1/		3.1	4.5.8.1/4.5.8.1.2
121	Hardcoat thickness test 1/		3.1	4.5.8.2
122	Microstructure 1/		3.7.1	4.5.6
123	Verification of aging	100%	3.7.2	4.5.1.6
124	Salt spray (chemical conversion coating) 1/		3.10.2	4.5.8.1/4.5.8.1.4

NOTES:
1. Sample size as specified in applicable test method paragraph.
3. Inspect prior to application of chemical conversion coating.
4. Applicable to M692/M731 only.
5. Applicable to M718A1/M741A1 only.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Base	SHEET 3 OF 3	9298356, 9277913 or 10542904 (as applicable)
			NEXT HIGHER ASSEMBLY: 10542902-1, -2 or -3 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Diameter of thread pilot 3/	Level V	3.1	Gage
202	Diameter of thread undercut 3/	Level V	3.1	Gage
203	Length, total	Level V	3.1	Gage
204	Length from front face to bottom of wrench holes, min	Level V	3.1	Gage
205	Width of thread undercut 3/	Level V	3.1	Gage
206	Presence and location of wrench holes	Level V	3.1	Gage
207	Radii or chamfers missing or incorrect	Level V	3.1	Visual/Func. Gage
208	Surface finish improper	Level V	3.1	Visual
209	Protective coating inadequate or damaged	Level V	3.1	Visual
210	Burr	Level V	3.11	Visual
211	Foreign matter	Level V	3.11	Visual
212	Hard anodic coating on threads, or indicated surface	Level V	3.1	Visual
213	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:
3. Inspect prior to application of chemical conversion coating.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Body (prior to banding)	1 OF 1	9298354, 10542912 or 9278078 (as applicable)
			NEXT HIGHER ASSEMBLY: 10542911-1, -2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Metal defective in band seat area	100%	3.5.2	Visual
Major				
101	Diameter of rotating band seat	Level III	3.1	Gage
102	Location of rotating band seat	Level III	3.9.2	4.5.2
103	Runout of band seat to bourrelets	Level III	3.1	Gage
Minor				
201	Radii or chamfers missing or incorrect	Level V	3.1	Visual
202	Burr	Level V	3.11	Visual
203	Foreign matter on band seat	Level V	3.3.3	Visual
204	Surface finish of band seat improper	Level V	3.1	Visual
205	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.6	TITLE / M483A1 / M718A1/M741A1 / M692/M731	SHEET 1 OF 5	DRAWING NUMBER See title block
	Body: 10542912 / 9278078 / 9298354		NEXT HIGHER ASSEMBLY 10542910-1, -2 or 9347119 (as applicable)
	Body Wrapped: 10542911-1 / 10542911-2 / 10542911-3		

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Metal defective	100%	3.5	4.5.3.2
Special				
a	Band seat metal defective 1/		3.3.1/ 3.5.2	4.5.3.1
Major				
101	Hardness test of body	100%	3.2	4.5.1.2
102	Yield strength and percent elongation 1/		3.2	4.5.1.4
103	Band chemical analysis 1/		3.1	4.5.4
104	Torque of fiberglass wrapping to body 1/		3.4	4.5.9
105	Fiberglass content 1/		3.1	4.5.10
106	Runout of body undercut with bourrelets before fiberglass wrap	Level III	3.1	Gage
107	Runout of forward body ogive with bourrelets	Level III	3.1	Gage
108	Runout of rear body ogive with bourrelets	Level III	3.1	Gage
109	Runout of rotating band lip with bourrelets	Level III	3.1	Gage
110	Runout of fiberglass wrapping with bourrelets	Level III	3.1	Gage

NOTES:
1. Sample size as specified in applicable test method paragraph.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE			SHEET	DRAWING NUMBER
4.4.2.6	M483A1	M718A1/M741A1	M692/M731	2 OF 5	See title block
	Body 10542912	9278078	9298354		NEXT HIGHER ASSEMBLY
	Body Wrapped 10542911-1	10542911-2	10542911-3		10542910-1, -2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
111	Runout of pitch diameter of base thread with bourrelets and rear face	Level III	3.1	Gage
112	Runout of rear outside diameter with bourrelets	Level III	3.1	Gage
113	Runout of body cavity with bourrelets	Level III	3.1	Gage
114	True position of body ogive thread counterbore with minor diameter of body ogive thread	Level III	3.1	Gage
115	Pitch diameter of body ogive thread, max	Level III	3.1	Gage
116	Minor diameter of body ogive thread, max	Level III	3.1	Gage
117	O.D. of undercut (before fiberglass)	Level III	3.1	Gage
118	Length to forward body ogive basic diameter	Level III	3.1	Gage
119	Length to rear body ogive basic diameter	Level III	3.1	Gage
120	Length total	Level III	3.1	Gage
121	Length from rear face to front of rear bourrelet	Level III	3.1	Gage
122	Length from rear face to rear of forward bourrelets	Level III	3.1	Gage
123	Location of forward cannelure groove	Level III	3.1	Gage
124	Location of rear cannelure groove	Level III	3.1	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	M483A1	M718A1/M741A1	M692/M731	SHEET 3 OF 5	DRAWING NUMBER
4.4.2.6	Body	10542912	9278078	9298354		See title block
	Body Wrapped	10542911-1	10542911-2	10542911-3		NEXT HIGHER ASSEMBLY: 10542910-1, -2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
125	Straightness of diametral element of rear face	Level III	3.1	Gage
126	O.D. of bourrelets, min (front & rear)	Level III	3.1	Gage
127	O.D. of fiberglass	Level III	3.1	Gage
128	O.D. of rotating band lip	Level III	3.1	Gage
129	O.D. of rotating band flats	Level III	3.1	Gage
130	O.D. of rotating band cannelures	Level III	3.1	Gage
131	O.D. of obturator seat	Level III	3.1	Gage
132	I.D. of body cavity 2/	Level III	3.1	Gage
133	O.D. rear of body	Level III	3.1	Gage
134	Depth of body cavity keyway 3/	Level III	3.1	Gage
135	Location to bottom of body cavity keyway 3/	Level III	3.1	Gage
136	Length of body cavity	Level III	3.1	Gage
137	Length of body cavity keyway from rear face	Level III	3.1	Gage
138	Length from rear face to rear of rotating band	Level III	3.1	Gage
139	Thickness of wall from bottom of keyway to outer surface, at forward end of body cavity	Level III	3.1	Gage
140	Thickness of wall at forward end of cavity	Level III	3.1	Gage
141	Width of rotating band	Level III	3.1	Gage

NOTES:
2. For M692/M731 examine beyond tapered section.
3. Alternates.

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PARAGRAPH	TITLE			SHEET	DRAWING NUMBER
4.4.2.6		M483A1	M718A1/M741A1 M692/M731	4 OF 5	See title block
	Body	10542912	9278078 9298354		NEXT HIGHER ASSEMBLY
	Body Wrapped	10542911-1	10542911-2 10542911-3		10542910-1, -2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
142	Width of rotating band relief groove	Level III	3.1	Gage
143	Width of body cavity keyway	Level III	3.1	Gage
144	Pitch diameter of base thread	Level III	3.1	Gage
145	Minor diameter of base thread, max	Level III	3.1	Gage
146	True position of base thread counterbore with pitch diameter of base thread	Level III	3.1	Gage
147	Length of effective base thread, min	Level IIi	3.1	Gage
148	Length from front of band to band basic diameter	Level III	3.1	Gage
149	Runout of projected pitch diameter of ogive thread to bourrelets and ogive mating surface (2 places) 7/	Level III	3.1	Gage
150	Runout of tapered section 4/	Level III	3.1	Gage
151	Location of rear I.D. basic diameter 4/	Level III	3.1	Gage
152	Distance between I.D. basic diameters 4/	Level III	3.2	Gage
153	O.D. of bourrelets, max 6/ 5/	100%	3.1	Ring Gage
154	Band seat visible in rotating band cannelures	100%	3.3.2	Visual
155	Surface finish improper	Level III	3.1	Visual

NOTES:
4. Applicable to M692/M731 only.
5. Not required when 100% gaging bourrelet diameter on projectile, metal parts (after paint).
6. Inspection to be performed on fiberglass wrapped body. Ring gage must pass over fiberglass wrap.
7. This inspection not required when components are to be supplied as assemblies.

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PARAGRAPH	TITLE	M483A1	M718A1/M741A1	M692/M731	SHEET 5 OF 5	DRAWING NUMBER
4.4.2.6	Body	10542912	9278078	9298354		See title block
	Body Wrapped	10542911-1	10542911-2	10542911-3		NEXT HIGHER ASSEMBLY: 10542910-1, -2 or 9347119 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Diameter of body ogive thread counterbore	Level V	3.1	Gage
202	Depth of body ogive thread counterbore	Level V	3.1	Gage
203	Diameter of base thread counterbore	Level V	3.1	Gage
204	Depth of base thread counterbore	Level V	3.1	Gage
205	Radii or chamfers missing or incorrect	Level V	3.1	Visual
206	Presence and size of undercut forward of rotating band (if applicable)	Level V	3.1	Visual/Gage
207	Burr	Level V	3.11	Visual
208	Foreign matter	Level V	3.11	Visual
209	Length to rear of fiberglass undercut, min	Level V	3.1	Gage
210	Length to front of fiberglass undercut, max	Level V	3.1	Gage
211	Protective coating inadequate or damaged	Level V	3.1	Visual
212	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:

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PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Projectile, Subassembly (before paint)	1 OF 1	10542910-1, -2 or 9347119 (as applicable)
			NEXT HIGHER ASSEMBLY: 10542902-1, -2 or -3 (as applicable)

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Metal defective stamp missing (indicating failure to pass test)	100%	3.1	Visual
b	Metal defective	100%	3.5	Visual (probe light)
Major				
101	Weight	Level III	3.1	Approved scale
102	Runout of projected pitch diameter of fuze thread with bourrelets and front face (2 places)	Level III	3.1	Gage
103	Runout forward of ogive with bourrelets	Level III	3.1	Gage
104	Runout rear of ogive with bourrelets	Level III	3.1	Gage
105	Gap between ogive and body	Level III	3.1	Gage
106	Length total, max	Level III	3.1	Gage
107	Air leak test 1/	100%	3.1	4.5.7
Minor				
201	Burr	Level V	3.11	Visual
202	Foreign matter	Level V	3.11	Visual
203	Protective coating inadequate or damaged	Level V	3.1	Visual
204	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:
1. Not applicable to M692/M731.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.8	TITLE Projectile, Metal Parts (after paint)	SHEET 1 OF 2	DRAWING NUMBER 10542902-1, -2 or -3 (as applicable) NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Special				
a	Ballistic test			
	a. Excess pressure 1/		3.6	4.5.12.1
	b. Zone 3 1/		3.6	4.5.12.2
	c. Service pressure 1/		3.6	4.5.12.3
b	Lifting plug missing	100%	3.1	Visual
Major				
101	Salt spray 1/		3.1	4.5.8.1
102	Diameter of bourrelets, max 2/	100%	3.1	Ring Gage
103	Rotating band damaged	Level III	3.1	Visual
104	Fiberglass wrapping damaged	Level III	3.1	Visual
105	Obturator missing or damaged	Level III	3.1	Visual
106	Threads damaged	Level III	3.11.3	Visual
107	Protective finish incomplete or damaged	Level III	3.11.4	Visual
108	Lifting plug detorque	Level III	3.1	Gage

NOTES:
1. Sample size as specified in applicable test method paragraph.
2. Ring gage must pass over the fiberglass wrapped section. At the contractor's option, the max diameter may be gaged at Level III inspection when 100% inspection (4.4.2.6, Major 153) of bourrelets is performed on the fiberglass wrapped body before paint.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.8	TITLE Projectile, Metal Parts (after paint)	SHEET 2 OF 2	DRAWING NUMBER 10542902-1, -2 or -3 (as applicable) NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Paint on rotating band bearing surfaces in excess of drawing allowance	Level V	3.1	Visual
202	Paint of threads	Level V	3.1	Visual
203	Paint on obturator	Level V	3.1	Visual
204	Paint not dry	Level V	3.1	Visual/Manual
205	Marking missing, incorrect or illegible	Level V	3.1	Visual
206	Burr	Level V	3.11	Visual
207	Foreign matter	Level V	3.11	Visual
208	Grommet missing or loose	Level V	3.1	Visual/Manual
209	VCI paper missing 3/	Level V	3.1	Visual/Manual
210	Protective caps (base thds) missing 3/	Level V	3.1	Visual
211	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:
3. Not required for intraplant use.

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QUALITY CONFORMANCE INSPECTION
CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.2.9	Pallet 1/	SHEET 1 OF 1	8837839
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major	None defined			
Minor				
201	Pallet damaged (unserviceable)	Level V	3.1	Visual
202	Improperly packed	Level V	3.1	Visual
203	Strapping missing, loose or broken	Level V	3.1	Visual/Manual
204	Marking missing, incorrect or illegible	Level V	3.1	Visual
205	Evidence of poor workmanship	Level V	3.11	Visual

NOTES:
1. Not required for intraplant use.

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4.4.3 Testing. Testing is described in the First Article Table, Quality Conformance Inspection Tables, and herein.

4.4.3.1 Hardness prior to cold forming (special defect). When the contractor elects to cold form the body he will assure compliance with 3.2.1. Bodies subjected to this test will be hardness tested at a midpoint along the length of outside diameter of the body. For a minimum of 20,000 bodies, each body will be tested. Based upon conformance with the requirement during this testing, the contractor may propose an alternate hardness sampling procedure. As a minimum, the alternate hardness sampling procedure shall include hardness testing of the first twenty consecutive bodies out of slow cool after each start up and sampling for hardness on a specific hourly basis. The alternate procedure shall be submitted to the procuring activity together with supporting hardness data for approval.

4.4.3.2 Verification of base heat treat (critical defect). After heat treat and prior to hardcoat, each base shall be verified for its having been heat treated. Bases failing to comply will be rejected. Rejected parts may be heat treated and resubmitted. All bases passing this test shall be stamped in accordance with applicable drawing. Testing shall be in accordance with 4.5.1.3.

4.4.3.3 Verification of base aging. After aging and prior to hardcoating, each base shall be verified for having been aged properly. Bases failing to comply will be rejected. All bases passing the test shall be stamped in accordance with applicable drawing. Testing shall be in accordance with 4.5.1.6.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Inspection Method Reference" column in the table starting with 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See section 6 of MIL-A-48078 and 6.4 herein.

4.4.4.1 Automatic inspection equipment. All automatic test and inspection equipment when utilized by the contractor shall be subjected to periodic verification or calibration check during production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/calibration. These procedures shall be included in the operation/calibration procedures as required by the above paragraph.

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4.4.4.2 Certification of nondestructive test (NDT) personnel. All personnel operating or calibrating NDT equipment shall be qualified and certified in accordance with MIL-STD-410.

4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Hardness prior to cold forming. Prior to testing, the surface of each body, in the area to be tested, shall be prepared by machining or grinding a flat of sufficient depth, .030 inches minimum, to remove scale and get below the decarburization zone. Test shall be performed using the procedures of ASTM Method E-103. Bodies failing to comply with 3.2.1 may be reannealed and retested.

4.5.1.2 Hardness (body). In order to determine uniformity of heat treatment, each body, after application of the welded overlay band and heat treatment and before final machining, shall be subjected to a hardness test with a Brinnel Hardness Tester or any other method approved by the procuring activity. Readings shall be taken on the body two and three quarters (2-3/4) inches forward of the finished rear face of body. This location must be appropriately increased to allow for the extra length of a rough turned body. The test shall be made in accordance with ASTM Method E-103 and the following:

a. The surface of the body in the area to be hardness tested shall be prepared by machining or grinding a flat of sufficient depth, 0.015 inches minimum, to remove scale and get below the decarburization zone. (Note: Flats and impressions shall be completely removed by final machining.)

b. Hardness tester periodic verification shall be performed using Brinnell test standards manufactured in accordance with ASTM Method E-10 in lieu of production samples.

c. All hardness test measurements shall be automatically recorded.

d. The hardness tester shall use a standard 10mm Brinnell ball. The hardness tester shall determine hardness in accordance with the following test sequence. First, the tester shall clamp the body to be tested with a 10,000-500 lb clamping load. Second, apply a minor load of approximately 10 percent of the 3000 kg of total test load. Third, apply the total test load. The clamping load will not be exceeded when the test load is applied. Fourth, relax the total test load back to the minor load. Fifth, automatically measure this penetrator travel from initial minor load position to new minor load position following application of the total test load. Sixth, display hardness and release the body.

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4.5.1.3 Verification of heat treat of base. The test shall be made using the procedures as prescribed in ASTM-E18. The readings shall be taken on the surface indicated on the applicable drawing.

4.5.1.4 Yield strength and percent elongation of body. The hardest and two (2) softest bodies from each standard heat treatment lot or the two (2) hardest and four (4) softest bodies from each miscellaneous heat treatment lot as determined by the hardness test in 4.5.1.2 shall be subjected to this test. Sample for mechanical property testing shall be selected after heat treatment of the entire heat treatment lot. Two tension test specimens conforming to the largest obtainable round type specimen as prescribed by ASTM E-8 shall be taken longitudinally from opposite sides of each body, midwall of the band seat area and two specimens as prescribed by ASTM E-8 shall be taken longitudinally from opposite sides of the body one inch from the front end. The number of specimens and their location are for the convenience of the contractor and Government and shall not restrict the contractor or Government from testing specimens at other locations to determine that the mechanical properties of the body meet the drawing requirements. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. At the option of the contractor, the entire lot may be reheat treated and retested, or a new set of bodies may be selected and tested from a segment of the heat treat lot having a tighter hardness range than the heat treatment lot as originally sampled. If an acceptable hardness range can be established for this heat treatment lot, bodies falling within this range will be accepted. All other bodies will be rejected. Rejected bodies may be reheat treated and retested.

4.5.1.5 Yield strength and percent elongation of ogive and base. The sample size for the tension test shall be one component (each base and ogive) from each 1200 parts or portion thereof from each heat treatment lot. Two longitudinal test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E-8 shall be taken from the sidewall of each sample. The base specimens shall be taken from midwall centered between rear and front ends 180° apart. The ogive tension test specimen shall be taken from the midwall at a location where the gage length center of the specimen is approximately one inch from the finished rear face of the body thread end of the ogive, 180° apart. The criteria for performing the tests shall be as prescribed by ASTM Method E-8. Failure of any specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the quantity represented by the sample.

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4.5.1.6 Verification of base aging test. The test shall be conducted using the procedures as prescribed in ASTM E1004. A minimum of three readings shall be taken on the surface indicated on the applicable drawing. Any part that fails to meet the minimum requirements of the applicable drawing will be rejected. Standards and test method shall be in accordance with drawing 9280406.

4.5.1.7 Stress corrosion crack inspection. One heat treated base forging shall be selected from the first production quantity and tested in accordance with the requirements of ASTM-G47. Three base forging specimens shall be extracted from the center of the base forging 1/2 inch from the front face. These specimens will be taken adjacent to each other. The specimens shall be stressed in tension at 35.0 ksi. Any specimen failing to comply with the requirements of ASTM B247 and the applicable drawing shall be cause for rejection.

4.5.2 Rotating band seat location. The contractor may use any control method that will assure the band seat meets the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal to the characteristics defined in 4.4.2.5. When the contractor establishes a control method, the manufacturing procedure, and inspection plan shall be forwarded to the Technical Agency for approval prior to use in accordance with 3.9.2. No change to these controls will be made once established without approval from the technical Agency.

4.5.3 Inspection for body metal defects after machining.

4.5.3.1 Inspection of band seat.

4.5.3.1.1 Welder qualification sampling. Initially, the first five banded bodies after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to the tests outlined below. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been found acceptable. These bodies shall have the band removed by machining and chemically etching without damaging the band seat base metal (steel) (see 6.8). These bodies will be subjected to the following tests:

a. Visual examination for compliance with 3.3.1.1 minimum melt requirements.

b. Visual examination for compliance with 3.5.2 metal defects, using the technique described in 4.5.3.1.3.

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c. Turn down band seat to the diameter specified in 3.3.1.2 and again visually inspect in accordance with 4.5.3.1.3. Any defect indication per 3.3.1.2 at this depth is considered to be a reject.

d. Should any sample for any of the above tests fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

4.5.3.1.2 Post qualification sampling.

4.5.3.1.2.1 Etched bandseat examination. Reduced sampling of one in each eight hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been found acceptable (see 6.10). One body per welder will be randomly selected and inspected according to the following criteria:

a. Visual examination for compliance with 3.3.1.1, minimum melting. The body selected for this test need not be the test sample having the lower rotating band iron content when tested in accordance with 4.5.4.

b. Visual inspection compliance with 3.5.2, metal defects, using the technique described in 4.5.3.1.3. If the sample fails to comply with 3.5.2 then the band seat will be turned down to the diameter specified in 3.3.1.2 and again visually inspected in accordance with 4.5.3.1.3.

c. Should any sample fail to comply with the minimum acceptance requirements, the quantity of bodies represented by that sample shall be rejected.

d. Two consecutive failures when tested for a, or two consecutive failures when tested for b, for any welding machine will mandate the return to the original welder qualification sampling of 4.5.3.1.1. However, reduced sampling of one in each eight hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

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4.5.3.1.2.2 Sub-surface defect examination. The body having the highest acceptable iron content rotating band when tested per 4.5.4 during this eight hour post qualification sampling shall be tested for compliance with 3.3.1.2. This body will have its band seat turned down to the diameter specified in 3.3.1.2 and visually inspected per 4.5.3.1.3. Failure of the body to comply with the specified requirements will require rejection of all bodies having bands welded by the same welder during the same sampling period. Failure of any body will also require that the body exhibiting the highest iron content from each welder will be similarly tested. Failure of any body will result in rejection of the quantity from each discrete welder represented by the sample.

Two consecutive failures on any welding machine will mandate the return to the original welder qualification sampling of 4.5.3.1.1. However, reduced sampling of one in each eight hours production or fraction thereof may be implemented after the first five banded bodies have been found acceptable.

4.5.3.1.3 Magnetic particle inspection. Magnetic particle inspection shall be as specified in MIL-STD-1949 for wet fluorescent method with indirect magnetization by the continuous method, initially after inducing a circular magnetic field, and again after inducing a longitudinal magnetic field.

4.5.3.1.4 Retest. If the sample selected in 4.5.3.1.2.2 fails to comply with the requirements, then a minimum of 10% of the shell from the failed welder may be randomly selected for retest. These parts will be tested in accordance with 4.5.3.1.2.2. Only the retest per welder is allowed. Failure of any part will mandate the rejection of the product represented by the sample.

4.5.3.2 Finished body. Projectile bodies after final machining and prior to fiberglass wrap shall be inspected by the magnetic particle inspection technique. At the option of the contractor, this test may be performed prior to machining either the keyway or threads. The procedures for performing this test shall be as specified in MIL-STD-1949, for wet fluorescent method with indirect magnetization by the continuous method. The entire body, inside and outside, shall be inspected after inducing a circular magnetic field and again after inducing a longitudinal magnetic field. The magnetic particle standard per Dwg. 9280318 shall be utilized to establish inspection sensitivity. Any body which fails to comply with the requirements of 3.5 shall be rejected. Bodies which comply shall be stamped with the letter "M" as specified on the applicable drawing.

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4.5.3.3 Magnetic particle suspension. The solid content of the suspension shall be maintained at the level recommended by the suspensoid manufacturer. The suspension shall be checked at the start of each day's production and at intervals of not over four hours of continuous operation, utilizing the procedures and equipment prescribed by section 4 of MIL-STD-1949. When performing this test for solid content of the suspension, if it is determined that it does not comply, all bodies that have been tested and accepted per 4.5.3.2 since the last satisfactory test was performed shall be retested after the suspension has been brought to the proper level. The magnetic particle test shall be performed on equipment that complies with the requirements of specification MIL-M-6867.

4.5.4 Band chemical analysis. The first five banded bodies, after the welded overlay band has been applied, for each welding machine in operation, shall be subjected to this test. Reduced sampling of one in twenty banded bodies may be implemented after five consecutive banded bodies per each welding machine in operation have been tested and found acceptable. Reduced sampling of one in each four hours production or fraction thereof per welding machine in operation may be implemented after five consecutive samples of one in twenty banded bodies per each welding machine in operation have been tested and found acceptable. Sample chips shall be obtained from each band from the last .050 inches removed when machining to the finished diameter. All chips shall be collected and identified with the shell from which the chips were machined. A minimum five gram sample from each complete specimen thus obtained from each band shall be analyzed by any method approved by the technical agency. The sample may not be cleaned with a magnet. Failure to comply with the requirements of the drawing shall be cause for rejection of the production represented by the sample.

4.5.5 Hydrostatic test of ogive. Prior to the application of protective finish, the finish machined ogive shall be subjected to a hydrostatic test. The ogive shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over four hours of continuous operation. It shall also be calibrated prior to re-starting the operation after a layoff of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the test pressure was actually lower than that specified on the applicable drawing, the ogives tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. The fixture should be such that it does not restrain the ogive. That is, it shall simulate actual assembly conditions.

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4.5.5.1 Lot sampling plan. One hundred percent testing of finish machined ogives shall be conducted until 1200 ogives consecutively pass the hydrostatic test. Subsequently lots of not greater than 5000 ogives may be submitted for testing. A sample of 50 ogives per lot shall then be tested. Failure of any ogive to meet the requirement of the applicable drawing shall be cause for rejection of the lot represented by the sample. When a lot has been rejected while on sampling inspection, the frequency shall revert back to the 100 percent inspection until 1200 ogives consecutively pass the hydrostatic test, to qualify for reduced sampling testing.

4.5.5.2 Continuous sampling (option). The ogive hydrostatic testing may be conducted using a continuous sampling plan with an i and f combination chosen from Table III below. One determining factor is the production internal size. Once the production rate is established, the contractor may choose from the combinations of clearance number (i) and corresponding frequency number (f) allowed by Table III. One hundred percent testing of machined ogives shall be conducted until i, consecutive ogives pass the hydrastatic test. Subsequently inspection continues at a fraction, f, of the units, where the sample units are selected in a random manner. Failure of any sample ogive to meet the requirement of the applicable drawing shall be cause for rejection of that sample and the reinstitution of 100 percent inspection. When i consecutive ogives pass the hydrostatic test, inspection at a frequency of f shall be resumed.

Table III: Applicable clearance number (i), frequency number (f) combinations for continuous sampling.

Number of units in production interval *	**Clearance Numbers i**							
26-90	194	321	420	498	NA	NA	NA	NA
91-500	194	321	420	498	623	NA	NA	NA
501-1200	194	321	420	498	623	762	NA	NA
1201-3200	194	321	420	498	623	762	930	NA
3201 and over	194	321	420	498	623	762	930	1147
	1/2	1/3	1/4	1/5	1/7	1/10	1/15	1/25
	Frequency Numbers f							

* A production interval is normally a shift; it can be a day if it is reasonably certain that shift changes do not affect quality of product, but shall not be longer than a day.

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4.5.6 Microstructure examination (ogive and base). One ogive and one base forging from each heat treatment lot shall be selected for microstructure examination. The metallographic specimen shall be etched with Keller's reagent and all areas examined under a microscope/metallograph at a minimum magnification of 250 diameters. Any evidence of the insoluble phase generally occurring in a semi-continuous or continuous network form shall be cause for rejection of the lot represented by the sample. Microstructural preparation and examination shall be performed by a metallographer or metallurgical engineer familiar with aluminum alloys.

4.5.6.1 Ogive specimen. A metallographic specimen at least 1/2 inch in length shall be removed from the sidewall. It shall be a longitudinal cross section of the body thread area with the center of the specimen approximately 3/4 inches from the finished rear face.

4.5.6.2 Base specimen. A metallographic specimen at least 1/2 inch in length shall be removed from the base and be a longitudinal cross section with its center located at mid-diameter of the base.

4.5.7 Air leak test. The ogive to body joint of the projectile subassembly shall be tested in accordance with the applicable drawing for 15 seconds minimum. A force shall not be applied in the axial direction such as to compress the ogive to body joint. Failure to comply shall be cause to reject the projectile subassembly. Projectile subassemblies which comply shall be stamped with the letter "A" in 1/8 inch letters in the area indicated on the drawing. The test may be accomplished by one of the following procedures.

4.5.7.1 Submersion test (alternate 1). Metal parts subassemblies shall be mounted in an approved fixture in such a manner that the subassembly shall be submerged in water. The required air pressure shall be applied and held for the minimum specified time, after the water in the tank is permitted to settle after immersion. Observation for leakage shall be made for the entire time the metal parts subassembly is required to be pressurized. The presence of air bubbles in the vicinity of the body to ogive joint or rising to the surface of the water shall be cause for rejection of the metal parts subassembly.

4.5.7.2 Pressure drop test (alternate 2). Each projectile subassembly shall be mounted so as to seal the projectile above and below the ogive to body joint with a collar (5 in^3 max.) and pressurizing this collar at 15 psi for 15 seconds while looking for a maximum pressure drop of 0.01 psi during the 15 second hold cycle. Pressure source shall be inactive during the hold cycle. A pressure drop in excess of 0.01 psi shall be cause for rejection of the subassembly.

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4.5.8 Protective finish test.

4.5.8.1 Salt spray test. The salt spray test shall be performed on each of the following: projectile subassembly (painted), ogive and base utilizing the equipment, methods and procedures specified in ASTM B-117 and in accordance with the applicable detailed specification for the protective finish, except the curing cycle for the body assembly shall be in accordance with component dwg. (M438A1-10542902; M718A1/M741A1-10542902; M692/M731-10542902) Testing shall be performed on one ogive and base for each twenty-four hours of continuous production or fraction thereof. The sampling for the subassembly shall be as specified in TT-C-490. When on reduced sampling (twice per week), one sample shall be taken during the first 30 minutes of production and a second taken during the last day of production that week. Failure to comply will be cause for rejection of the production since the previous test.

Note: The salt spray testing of the base's hard anodic coating and chemical conversion coating are permitted on the same test specimen.

4.5.8.1.1 Ogive salt spray test. The significant surface prescribed in MIL-A-8625 may be marked by tracing the outline of a 5 inch by 6 inch sheet of paper onto the ogive.

4.5.8.1.2 Base salt spray test for hard anodic coating. Bases shall be selected in accordance with 4.5.8.1. The significant surface specified in MIL-A-8625 shall be the flat surface of the cavity exclusive of the wrench holes.

4.5.8.1.3 Base salt spray test, nickel acetate seal. If the optional nickel acetate seal is used in lieu of a sodium dichromate seal or a duplex seal, five (5) of the ten (10) first article samples are to be subjected to a 1000 hours salt spray test. These samples must be produced on regular production equipment.

4.5.8.1.4 Base salt spray test for chemical conversion coating. Bases shall be selected in accordance with 4.5.8.1 in lieu of panels specified in MIL-C-5541. The test surface shall be the base threads and shoulder area (see 3.10.2).

4.5.8.2 Bases. One hard coated base per batch per tank shall be subjected to the thickness test prescribed in the applicable specification referenced on the drawing. Failure to comply shall be cause for rejection of the production since the previous test. Procedure must be approved by the technical agency.

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4.5.9 Torque of fiberglass wrapping to body. At a minimum one fiberglass wrapped body from each four hour's production from each spindle shall be subjected to this test to determine compliance with 3.4. After 10,000 fiberglass wrapped bodies have been produced and accepted, a reduced sample of one body from each 24 hours of continuous production or fraction thereof from each spindle in operation shall be subjected to this test to determine compliance with 3.4. The fiberglass wrapped body shall be mounted in a suitable holding device in such a manner that the body is prevented from turning. The fiberglass wrapping shall be notched to a suitable depth in as many places as necessary, 180 degrees apart. There shall be no compressive force when applying the applicable torque to the fiberglass wrapping. If any body fails to comply with the requirement, the production represented by the sample shall be rejected. An alternate test method and sampling plan may be submitted to the technical agency for approval.

4.5.10 Fiberglass content. The fiberglass wrapped bodies selected for the torque test shall also be subjected to this test. Three specimens (1 in. x 1 in.) shall be removed from the test shell. The specimens shall be individually weighed on an analytical balance. The specimens shall then be placed in a furnace at 1200°F for a period of 30 minutes. The specimens shall then be cooled in a desiccator and weighed. The heating and cooling procedure shall be repeated until the specimens attain a constant weight. The glass content is calculated by dividing the final weight by the initial weight and multiplying the result by 100. Failure of any specimen to comply shall be cause for rejection of the production represented by the sample.

4.5.11 Eddy Current inspection.

4.5.11.1 Eddy Current inspection of base. The base, prior to machining the holes, shall be inspected for metal defects using the Eddy Current method. Bases failing to comply with requirements of 3.5 shall be rejected. Standards and test method shall be in accordance with applicable drawing: a) M483A1-9280404, b) M718A1/M741A1-9280388, c) M692/M731-9280404.

4.5.11.2 Eddy Current inspection of ogive. The ogive, after machining, shall be inspected for metal defects using the Eddy Current method. Ogives failing to comply with the requirements of 3.5 shall be rejected. Each ogive passing this test shall be stamped in accordance with the applicable drawing. Standards and test method shall be in accordance with applicable drawing: a) M483A1-9280401, b) M718A1/M741A1-9280401, c) M692/M731-9280401.

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4.5.12 Proving ground ballistic tests. The following tests shall be conducted using cannon tubes having 25 percent minimum remaining life based upon current gun tube condemnation criteria. Photographic equipment shall be positioned near the muzzle of the weapon to record the condition of the projectile as it exits the tube. Chamber pressures and velocities shall be recorded for each round fired. All photographs and recovered hardware shall be examined for any noncompliance with the applicable requirements of 3.6. Range, deflection, and velocity shall be recorded for each round fired for information.

An attempt will be made to recover those projectiles fired per 4.5.12.1 only. A recovery shall be attempted any time an unusual firing occurs in any zone. Examples of unusual firings include, but are not limited to, erratic flight, projectile outside round impact area, fuze malfunction, or indication of metal parts separation.

4.5.12.1 Excess pressure. Each projectile submitted for this test shall contain inert cargo, have its base assembled, a dummy expulsion charge, and a dummy fuze. Each projectile will be fired with a PXR6297 propelling charge temperature conditioned to obtain a 47-51 thousand PSI chamber pressure and with a M199 cannon tube.

4.5.12.2 Zone 3. Each projectile submitted for this test shall contain inert cargo, have its base assembled, contain a live expulsion charge, and a M577 fuze. Each projectile will be fired in either an M185 or M199 cannon tube with a M3A1 propelling charge. The projectile complete with expulsion charge shall be conditioned to -40° + 5°F. Fuze function time will be recorded.

4.5.12.3 Service pressure. Each projectile submitted for this test shall contain inert cargo, contain a live expulsion charge, a M577 fuze. Each projectile will be fired in M185 or M199 cannon tube with an M119 series propelling charge at ambient conditions. Fuze function time will be recorded.

4.5.12.4 Sampling procedure. Normal sampling will be conducted until 5 consecutive lots have been accepted. Reduced sampling shall then be instituted. Any failure, while on reduced sampling, mandates return to normal sampling. Sample sizes for normal and reduced sampling shall be as follows:

	Normal	Reduced
a. Excess pressure	5	2
b. Zone three	3	3
c. Service pressure	2	-

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5. PACKAGING

5.1 Packing. This section (section 5) is not applicable if LAP operations are to be performed at the same facility.

5.2 Packaging, packing, and marking.

5.2.1 Projectile metal parts. Projectile metal parts assemblies shall be palletized in accordance with Drawing 8837839.

5.2.2 Bases. Bases packaging, packing and marking shall comply with the following:

5.2.2.1 Overseas shipments. Bases shall be single item packaged, moisture proof packaged (barrier bag only), packed, marked, and palletized in accordance with MIL-STD-1169, Level A.

5.2.2.2 Domestic shipments. Bases shall be separated or partitioned (if bulk) in the unit package to protect the surfaces and shall include marking to assure that the base is shipped or stored with the cavity facing downward. Palletization is optional.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile metal parts covered by this specification are intended for the Projectile, 155MM: M483A1, M718A1/M741A1 or M692/M731.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Applicable stock number.

c. First article sample requirements.

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract, per the Contract Data Requirements List, DD Form 1423.

Paragraph	Data Requirement	Applicable DID
3.2.1, 3.3.1, 3.9.2, 3.10, 4.5.2	Process Control/ Procedure Report	See DD Form 1423 in the contract

6.4 Submission of contractor inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: SMCAR-QAR-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.5 Inspection lot size. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.6 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards shall be forwarded to Commander, ARDEC, ATTN: SMCAR-QAR-R, and SMCAR-FSA-MS, Picatinny Arsenal, NJ 07806-5000, and AMSMC-QAD-D Army Master Depository, Rock Island, IL 61299-6000.

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6.7 Cold forming. The contractor is cautioned that if cold forming is utilized in processing of the projectile body, those bodies which do not completely clean up in the machining operation prior to the cold forming have a greater propensity to develop circumferential tears during the forming operation particularly if areas of non cleanup are heavily scaled.

6.8 Suggested etching procedures for removing gilding metal overlay.

a. Recommended. Immerse body in commercial copper stripper which is cyanide and chromate free and prepared in accordance with manufacturer's directions. In a fresh solution, an overlay layer 0.010 to 0.015 inch thick will dissolve in 10 to 14 hours. Renew solution as necessary.

b. Solution - 500 grams Chromic oxide Cr O_3 (commercial flakes) 50 ml. sulphuric acid H_2SO_4 (technical grade) 950 ml. water H_2O.

This solution may be used by passing current through the solution (50 amperes dc) with the immersed shell as the anode and the container as the cathode. An overlay layer 0.010 to 0.015 inch thick will require approximately two to three hours to dissolve in a fresh solution. The solution can also be used without an external current source by immersing the shell for approximately 8 to 12 hours. An active solution would clean all pits.

6.9 Alternate alloys. The specified alloys, 1340 and 4140, which are compatible with the welded overlay rotating band, have been successfully used as body material. Experience to date has indicated that 1340 steel requires closer control over the welding process and thermal treatment to prevent cracks after banding. However, this does not preclude the need for similar close control over the welding process and thermal treatment with 4140 steel to be compatible with the production process.

6.10 Band welding. Recommend welding operation of overlay welded band be guided by Frankford Arsenal "Process Recommendations for Banding 155MM, M483 Projectile" dated 19 April 1973. Requests for copies should be submitted to: Commander, ARDEC, ATTN: SMCAR-CCH-P, Picatinny Arsenal, New Jersey 07806-5000.

6.11 Steel blooms, billets, slabs and bars.

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6.11.1 Steel grade. It is the responsibility of the manufacturer of the projectile metal parts to select a steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operations of the shops of the forger and machiner to produce the projectile to the dimensions, mechanical properties and metal integrity specified as an end item requirement.

6.11.2 Raw material. The supplier of the selected steel raw material is responsible only for furnishing a metal composition, grade, and type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade (selected by the projectile manufacturer) and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

6.11.3 Forging quality processing techniques. In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimensions, in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices, and processing techniques having the capabilities of compensating for such variables.

6.12 Adequate working of ogive and base material. In order to insure adequate working of the ogive and base material and the break-up of insolubles from the cast ingot to the final raw forging configuration, it is recommended that a minimum extrusion ratio of six (6) be used representing an 83 percent reduction of area from the cross-sectional area of the extrusion.

6.13 Preproduction sample/hardness range. To minimize the need for reheat treating material or for having miscellaneous heat treatment lots it is recommended that the contractor process a preproduction sample to establish what should be an acceptable hardness range for each mill heat. This hardness range could then be used for screening bodies during processing of the heat treatment lot from that mill heat. When using a hardness range acceptance will still be in accordance with 4.5.1.4.

6.14 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristics may be measured or gaged to determine acceptability.

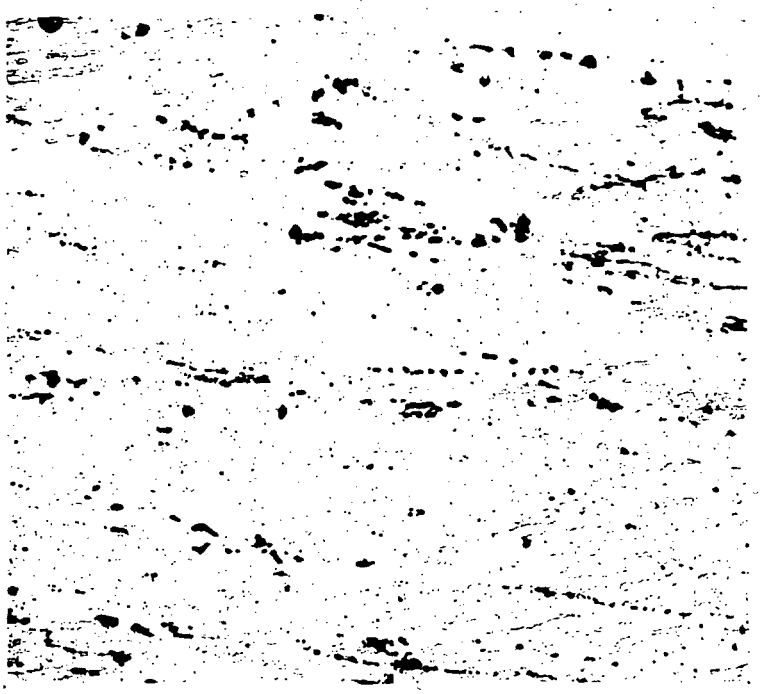
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6.15 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying alternative procedures, the contractor shall describe them in a written proposal and shall demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better than the quality assurance provisions specified herein.

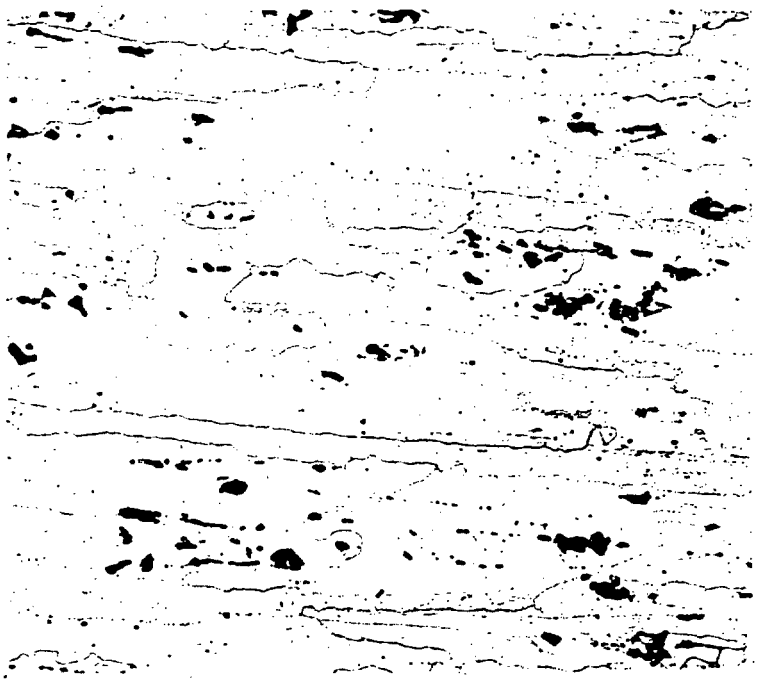
6.16 Drawings. Drawings listed in section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, U.S. Army Research and Development Command (ARRADCOM) or U.S. Army Armament Research and Development Center (ARDC) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.17 Submission of alternative quality conformance provision. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation by the technical activity responsible for the preparation of this specification.

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a



b

FIGURE 1 Acceptable Microstructures (250x)

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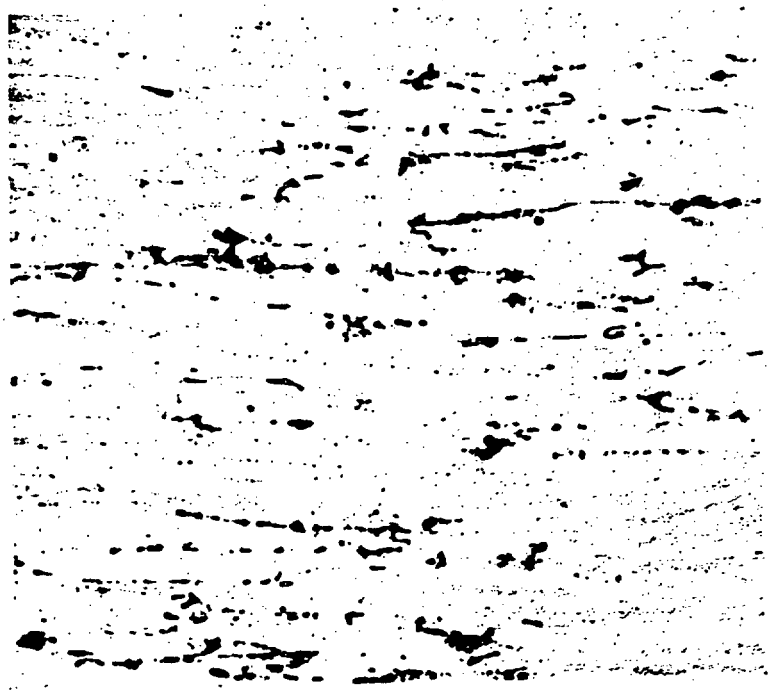
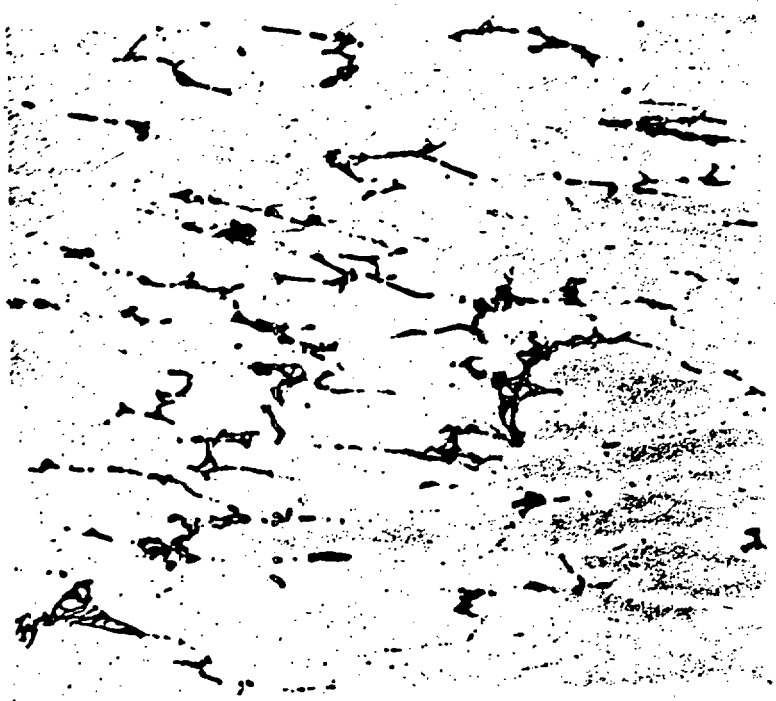


FIGURE 2 Marginally Acceptable Microstructure (250X)



b

FIGURE 3 Unacceptable Microstructure (250X)

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6.18 Subject term (key word) listing.

Air leak test
Ammunition
Ballistic testing
Eddy Current inspection
Hydrostatic test
Magnetic particle inspection
Microstructure
Stress corrosion crack

6.19 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-AD01)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-50345 (AR)	941004

3. DOCUMENT TITLE: PROJECTILES, 155MM, M483A1 - M718A1/M741A1 - M692/M731 METAL PARTS ASSEMBLY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S ARMY ARDEC STANDARDIZATION OFFICE	(1) Commercial 201-724-6675	(2) AUTOVON DSN-880-6675
c. ADDRESS (Include Zip Code) ATTN: AMSTA-AR-EDE-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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